

Fine-tuning, Instruction Tuning, and Midtraining

(Or, Training Beyond Pretraining)

Aaron Mueller

CS599 B1: Advanced Natural Language Processing

Sep. 24, 2026

Administrivia

- Reminder: **HW1** is due tonight at 11:59pm!
- **HW2** has been released! This will be due **Oct. 15**

Review: Training Transformers

So we now have our nearly optimally trained Transformer. Let's ask it some questions:

At what U.S. university was a woman first awarded a Ph.D.?

A. Vassar College B. Harvard University C. University of Pennsylvania D. Cornell University

Where can I find information on how to build a house from scratch?

Where can I find information on how to build a house from scratch?
Where can I find information on

Please rewrite the following passage in German: "What is the law of large numbers?" (you can assume that the person you are writing to is familiar with the concept of probability and statistics)

Hmm, not great. The model is acting like it was the one that posed the question!

How can we get models to be better at specific tasks?

Using an LM for a Specific Task

- We've seen examples of prompting LMs to do a specific task
 - Pros: Fast iteration, easy to implement
 - Cons: Often doesn't perform well, very finicky
- It's common to instead update the model's parameters for specific task
 - This is called **fine-tuning**

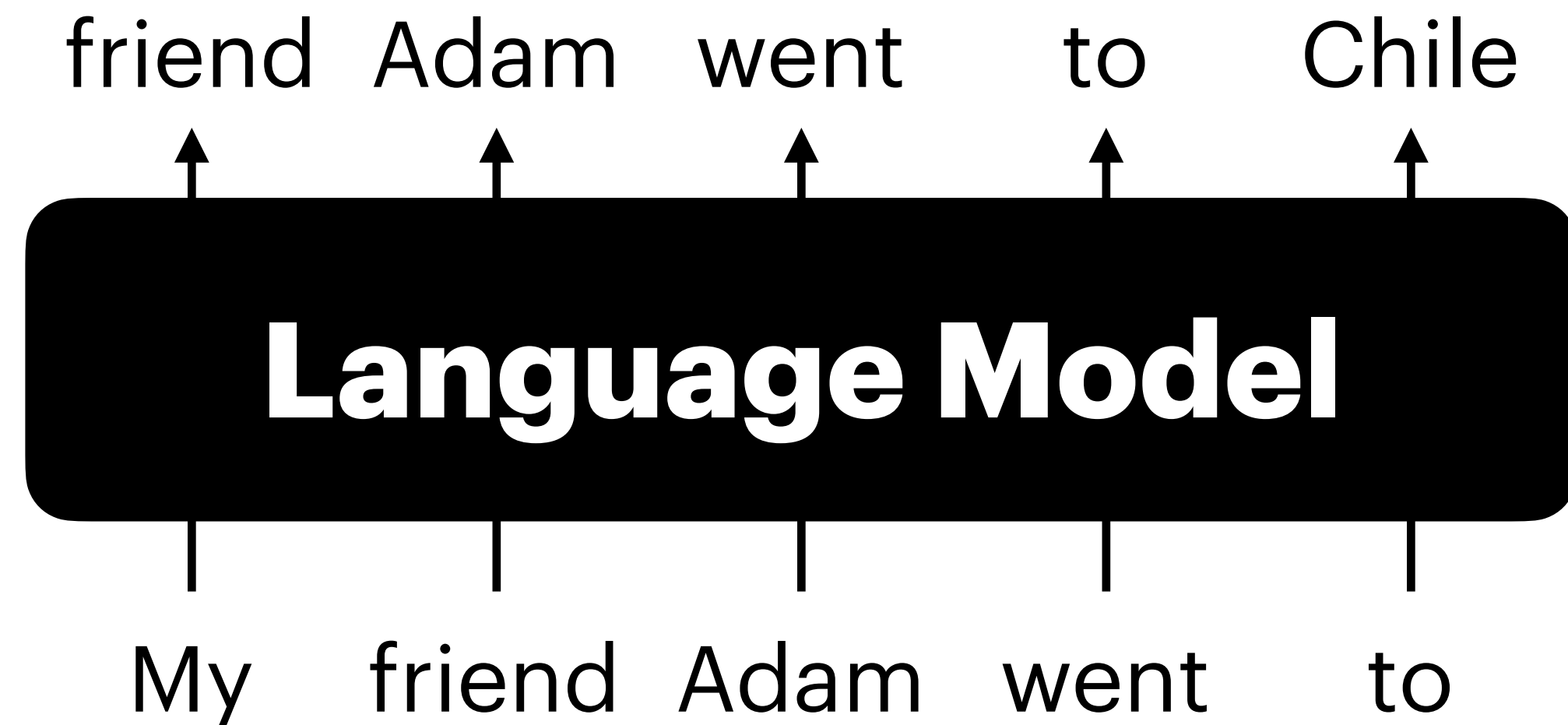
Fine-tuning

Unlike prompting, fine-tuning entails performing updates to the model's parameters to make it better at a specific task.

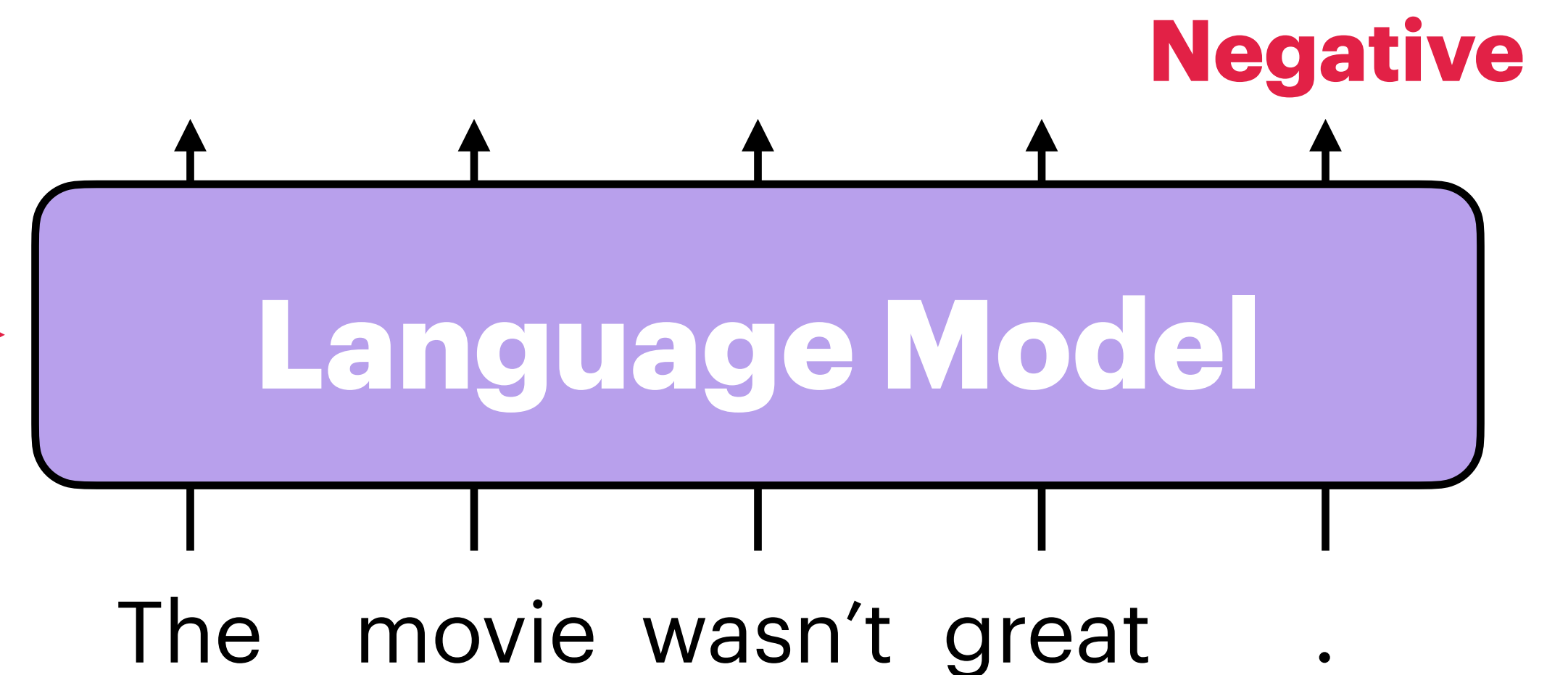


Fine-tuning

Stage 1: Pretrain

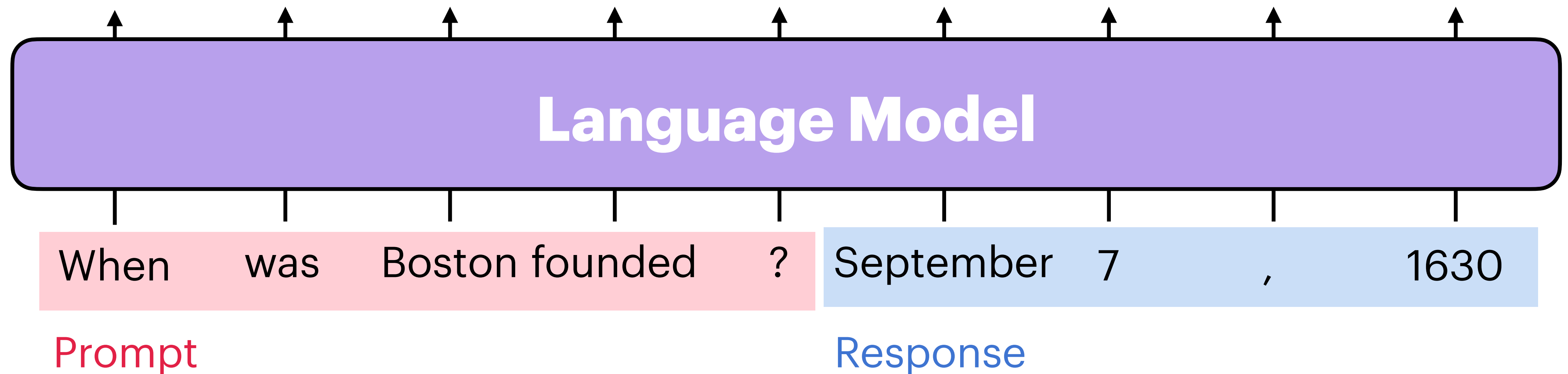


Stage 2: Fine-tune



We start with our pretrained model, and then update it for a specific task or domain.

Fine-tuning



It's similar to pre-training, except we only compute losses on the response. We want models to learn how to do a task without forgetting general language modeling.

Defining Fine-tuning

Given a batch of examples b : $\{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots\}$ from our dataset D , we want to take our pretrained model's parameters θ_p , and update them via gradient descent to reduce the loss:

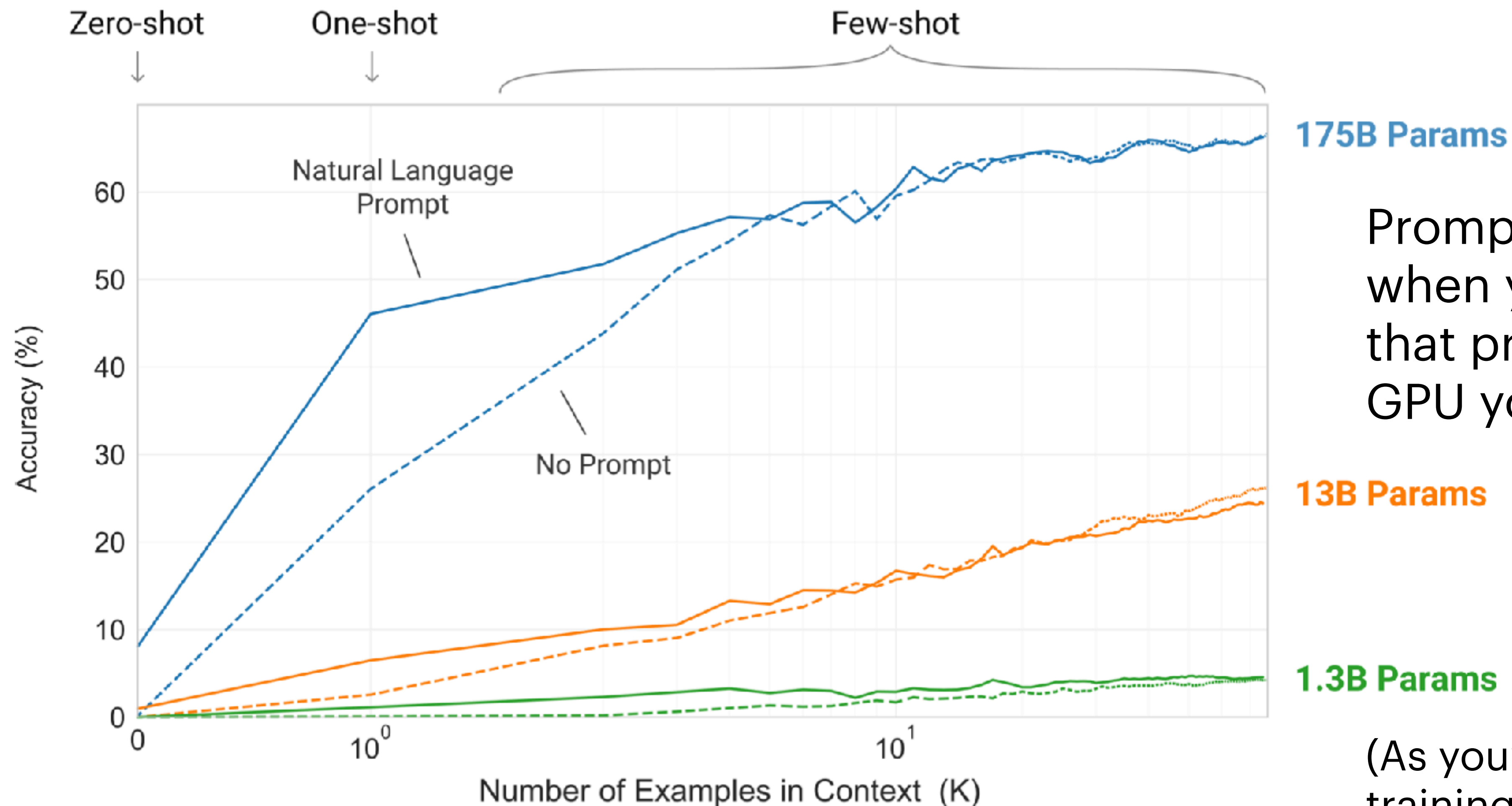
$$\theta_0 = \theta_p$$

$$\theta_i = \theta_{i-1} - \alpha \nabla_{\theta} \mathbb{E}_b[-\log p_{\theta}(y_i | \mathbf{x}_i)]$$

Same **cross-entropy loss**
from before—just new data!

So the main thing separating pre-training from fine-tuning is the data: pre-training is general, fine-tuning is (usually) task-specific.

Why fine-tune at all?



Prompting only works well when you have **huge** models that probably won't fit on any GPU you could keep at home.

(As you'll see in HW2, though, post-training helps smaller models a lot!)

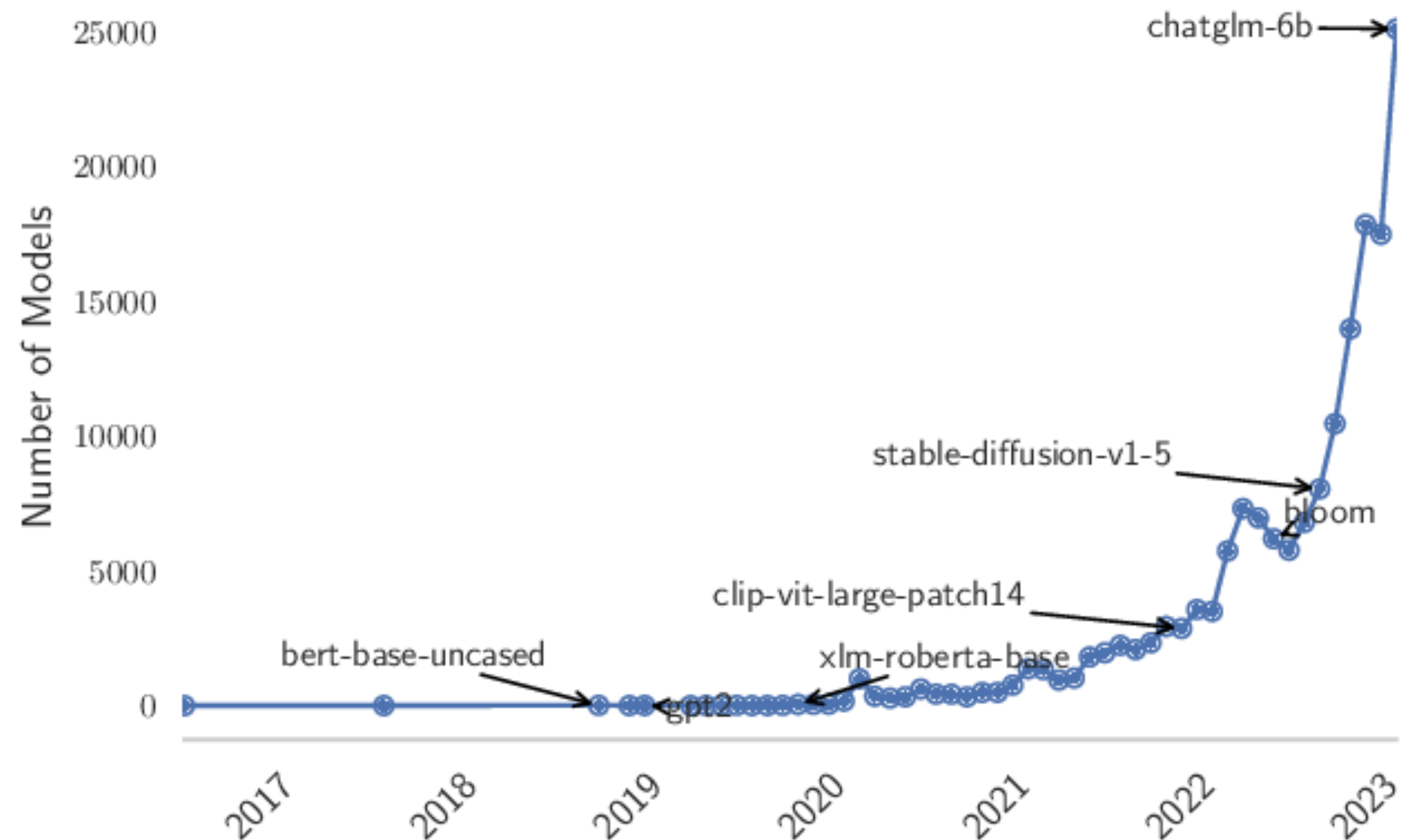
The Language Model Ecosystem



Hugging Face

There are tons of great models out there that you **can** download and fine-tune yourself!

- + Total control over your data
- + Can often get performance almost as good as proprietary models
- Your fine-tuned model will be more task-/domain-specific
- More and more SOTA results come from prompting proprietary models these days



The Performance of Fine-tuning

You can usually get comparable or better results by fine-tuning a smaller model on a large dataset than prompting a larger model. (Note: this is becoming less true by the day.)

	SuperGLUE Average	BoolQ Accuracy	CB Accuracy	CB F1	COPA Accuracy	RTE Accuracy
Fine-tuned SOTA	89.0	91.0	96.9	93.9	94.8	92.5
Fine-tuned BERT-Large	69.0	77.4	83.6	75.7	70.6	71.7
GPT-3 Few-Shot	71.8	76.4	75.6	52.0	92.0	69.0

	WiC Accuracy	WSC Accuracy	MultiRC Accuracy	MultiRC F1a	ReCoRD Accuracy	ReCoRD F1
Fine-tuned SOTA	76.1	93.8	62.3	88.2	92.5	93.3
Fine-tuned BERT-Large	69.6	64.6	24.1	70.0	71.3	72.0
GPT-3 Few-Shot	49.4	80.1	30.5	75.4	90.2	91.1

[Brown et al., 2020]

Why does fine-tuning work?

Why does this work? In theory, it shouldn't matter whether we start at θ_p or with some random initialization $\theta_0 \sim N$.

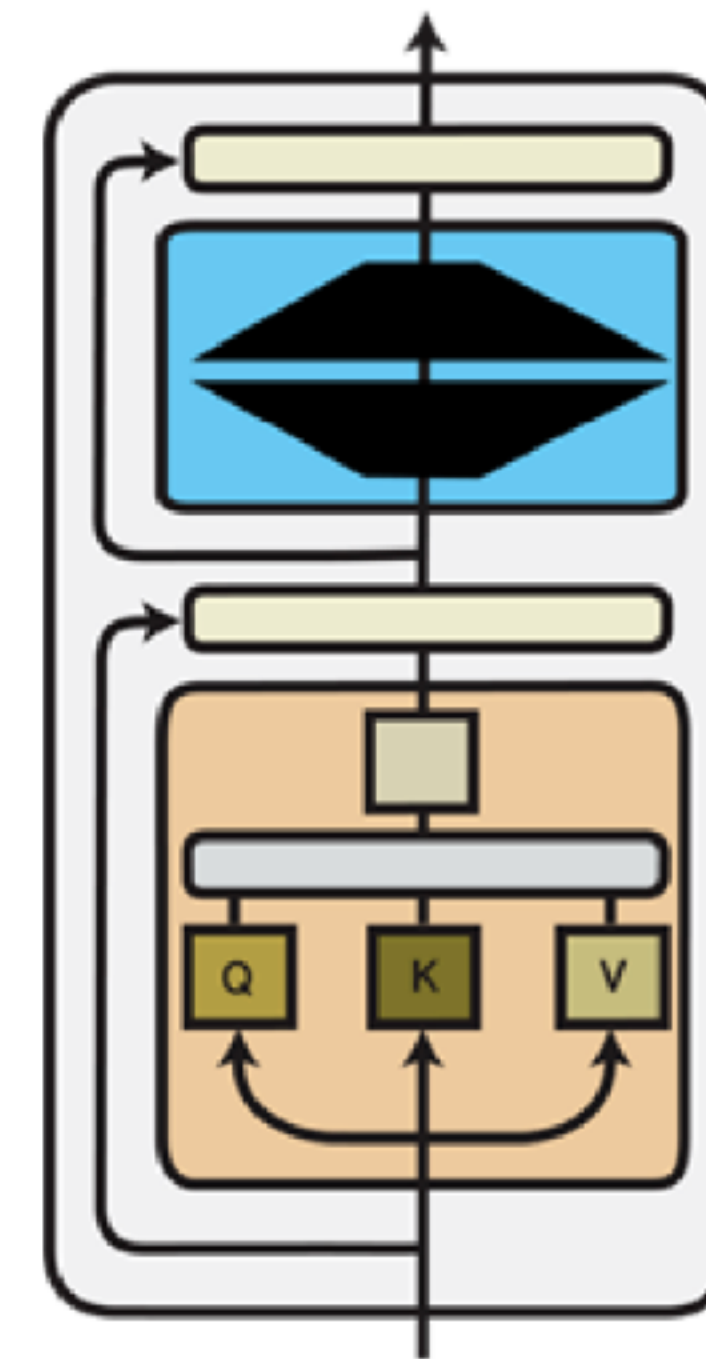
$$\theta_0 = \theta_p$$

$$\theta_i = \theta_{i-1} - \alpha \nabla_{\theta} \mathbb{E}_b[-\log p_{\theta}(y_i | \mathbf{x}_i)]$$

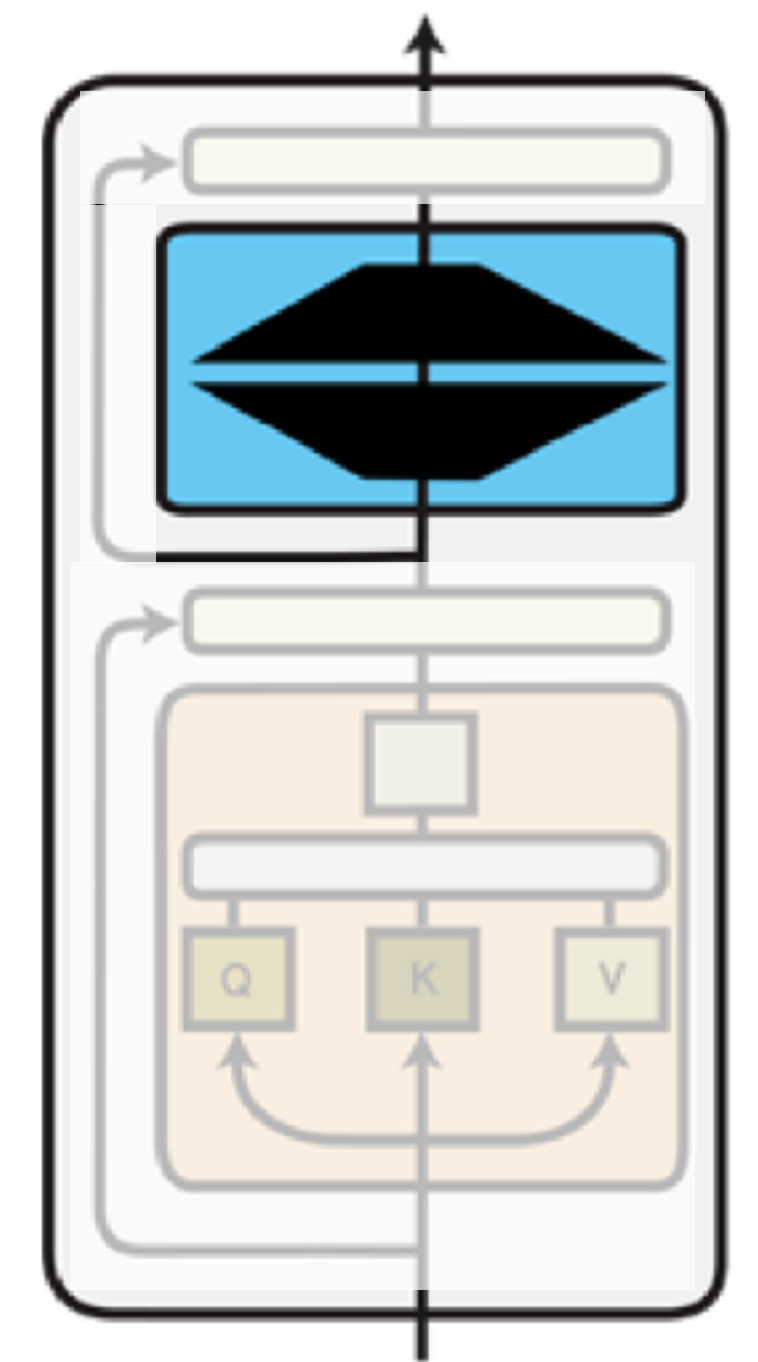
1. Pretraining teaches the model non-trivial concepts that it couldn't have learned from scratch from a more narrow distribution.
2. During fine-tuning, we don't stray too far from θ_p .

Parameter-efficient Fine-tuning

- There are a lot of ways to not stray too far from the pretraining parameters:
 - Small learning rate during fine-tuning
 - *Fine-tune only a subset of the parameters*
 - *Regularization terms*
- There are a lot of **parameter-efficient fine-tuning** methods that aim only to change the most important parameters.



Full fine-tuning



Parameter-efficient
fine-tuning

Low-rank Adaptation (LoRA)

Hu et al., 2021

- Introduce a learnable low-rank matrix as an update to the pretrained parameters:

$$\theta_1 = \theta_p + uv^\top \quad u \in \mathbb{R}^d \quad v \in \mathbb{R}^d$$

- This allows the parameters to change along a small number of directions.
- The above is a rank-1 example. Here's the generalization to rank- k :

$$\theta_1 = \theta_p + \sum_{j=1}^k u^{(j)} v^{(j)\top}$$

Like the above procedure applied k times!

- We minimize the loss by learning u and v :

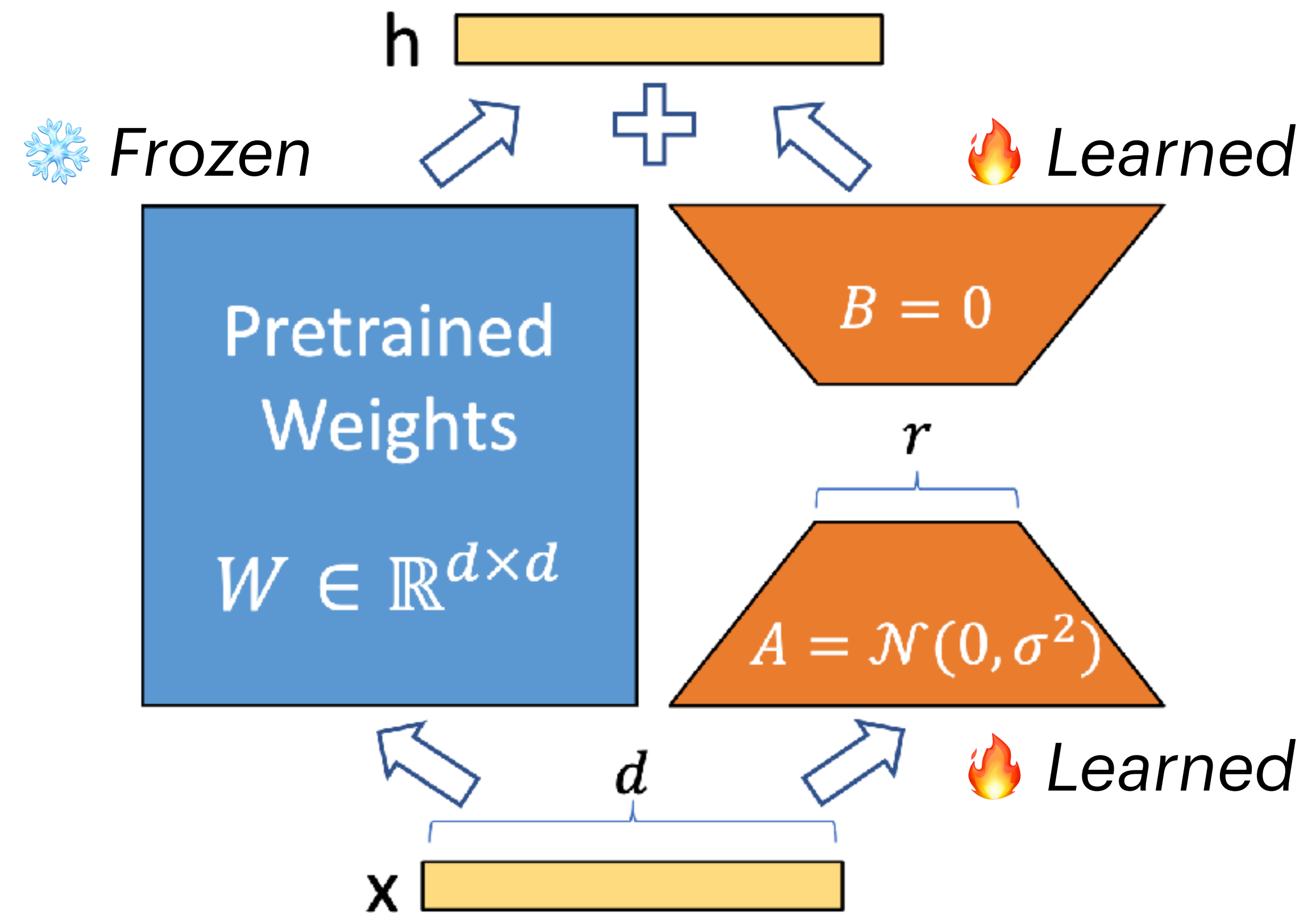
$$\min_{u_1, v_1, \dots, u_k, v_k} \mathbb{E}_{(x, y) \sim D} [L(x, y)]$$

Low-rank Adaptation (LoRA)

We can use matrices of size $A \in \mathbb{R}^{d \times r}$ and $B \in \mathbb{R}^{r \times d}$ to get:

$$\mathbf{h} = \mathbf{x}W + \mathbf{x}AB$$

- + Don't need gradients for most parameters: way lower memory usage
- + Doesn't add any time during inference
- + Modular: can swap out different LoRAs for different tasks/domains
- Quite a few more hyperparameters to fiddle with



[Hu et al., 2021]

Implementing LoRA

```
input_dim = 768
output_dim = 768
rank = 8
```

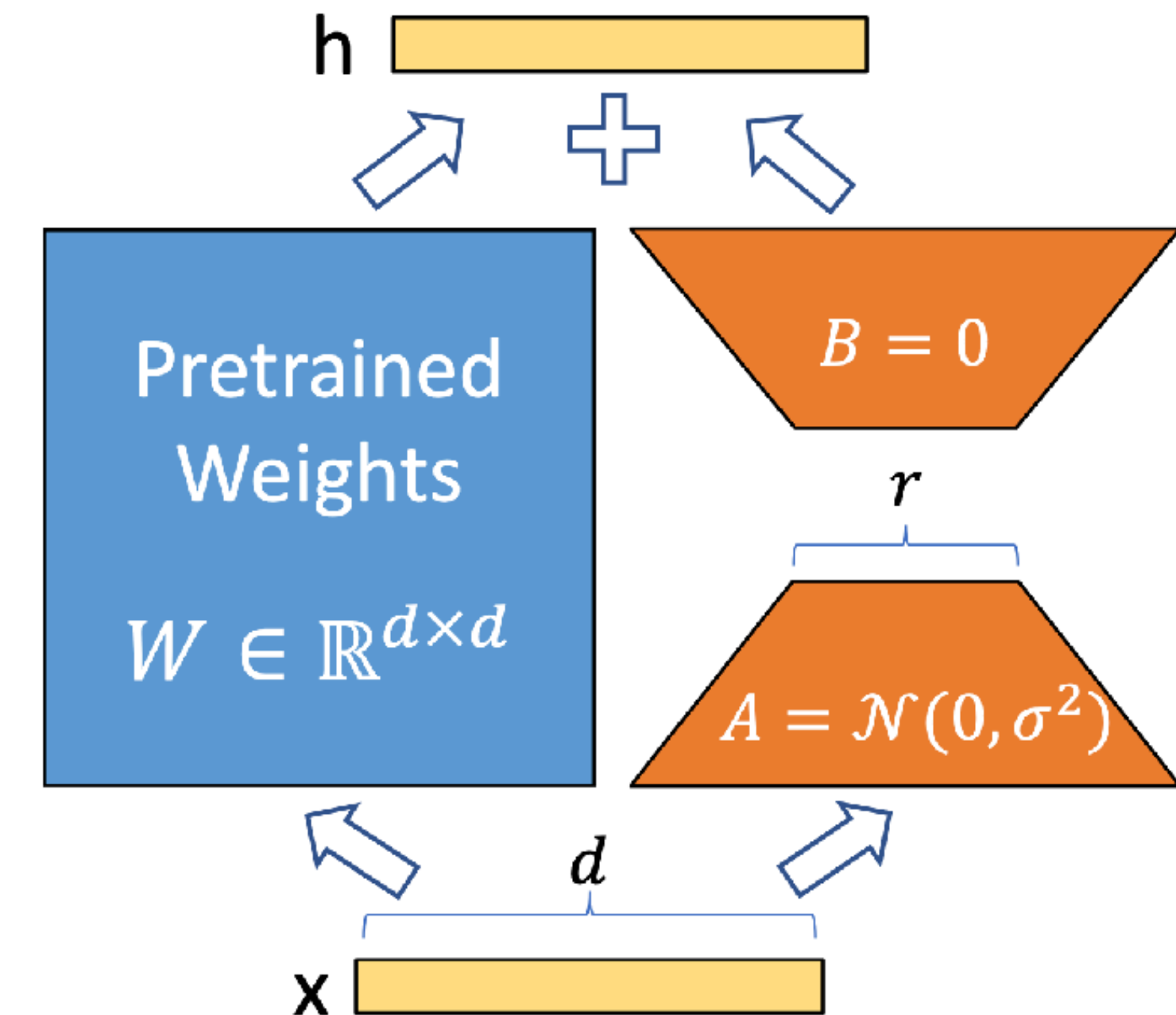
```
W = . . . # depends on pretrained model shape
```

```
W_A = nn.Parameter(torch.empty(input_dim, rank))
W_B = nn.Parameter(torch.empty(rank, output_dim))
```

```
nn.init.kaiming_uniform_(W_A, a=math.sqrt(5))
nn.init.zeros_(W_B)
```

```
def regular_forward_matmul(x, W):
    h = x @ W
    return h
```

```
def lora_forward_matmul(x, W, W_A, W_B):
    h = x @ W
    h += x @ (W_A @ W_B) * alpha
    return h
```



$$\mathbf{h} = \mathbf{x}W + \alpha \cdot \mathbf{x}AB$$

Controls strength of LoRA update

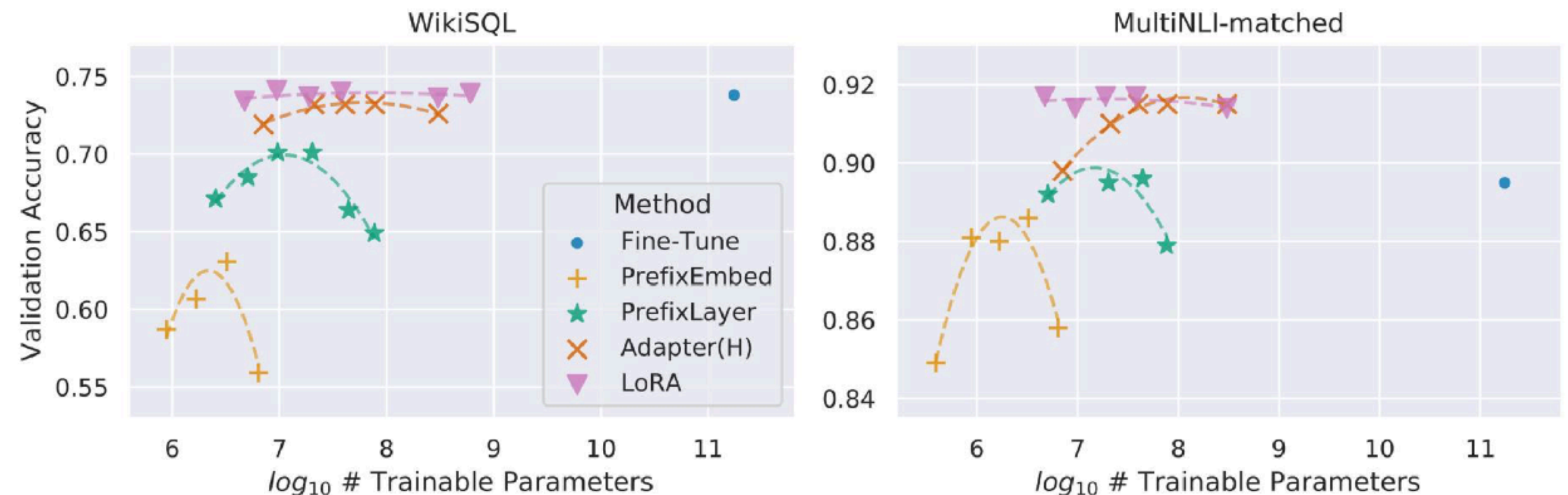
LoRA in Practice

LoRA performs just as well as full fine-tuning on a few NLP tasks.

FT = fine-tune
all parameters

Model&Method	# Trainable Parameters	WikiSQL	MNLI-m	SAMSum
		Acc. (%)	Acc. (%)	R1/R2/RL
GPT-3 (FT)	175,255.8M	73.8	89.5	52.0/28.0/44.5
GPT-3 (BitFit)	14.2M	71.3	91.0	51.3/27.4/43.5
GPT-3 (PreEmbed)	3.2M	63.1	88.6	48.3/24.2/40.5
GPT-3 (PreLayer)	20.2M	70.1	89.5	50.8/27.3/43.5
GPT-3 (Adapter ^H)	7.1M	71.9	89.8	53.0/28.9/44.8
GPT-3 (Adapter ^H)	40.1M	73.2	91.5	53.2/29.0/45.1
GPT-3 (LoRA)	4.7M	73.4	91.7	53.8/29.8/45.9
GPT-3 (LoRA)	37.7M	74.0	91.6	53.4/29.2/45.1

LoRA gives better performance at the same number of trainable parameters.



Regularization

We can add a **regularizer** to prevent the fine-tuned model from diverging too much from the pretrained model:

$$R = \|\theta_p - \theta_i\|_2$$

L_2 distance: how much have the parameters moved?

$$R = \text{KL}(p_{\theta_p}(\cdot | x_{<t}) \| p_{\theta_i}(\cdot | x_{<t}))$$

/

KL divergence: how much do the pretrained model and fine-tuned models' probability distributions differ?

$$\text{KL}(p \| q) = \sum p(x) \cdot \left(\log\left(\frac{p(x)}{q(x)}\right) \right)$$

Regularization

We can add a **regularizer** to prevent the fine-tuned model from diverging too much from the pretrained model:

$$R = \|\theta_p - \theta_i\|_2$$

L_2 distance: how much have the parameters moved?

$$R = \text{KL}(p_{\theta_p}(\cdot | x_{<t}) \| p_{\theta_i}(\cdot | x_{<t}))$$

/

KL divergence: how much do the pretrained model and fine-tuned models' probability distributions differ?

$$L = \underbrace{\mathbb{E}_b[-\log p_{\theta_i}(y_t | x_{<t})]}_{\text{cross-entropy}} + \underbrace{\mathbb{E}_b[\text{KL}(p_{\theta_p}(\cdot | x_{<t}) \| p_{\theta_i}(\cdot | x_{<t}))]}_{\text{regularization term}}$$

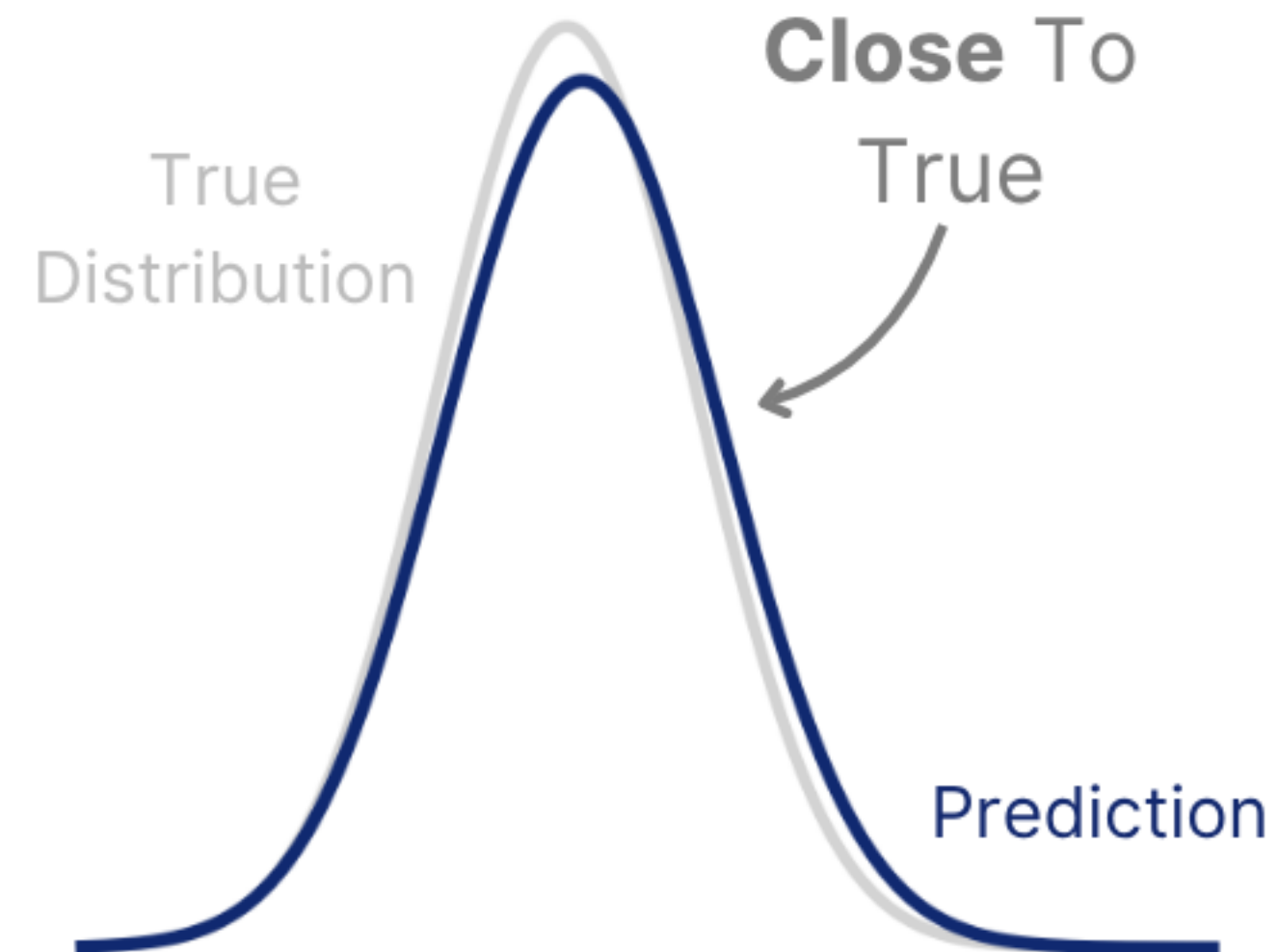
The KL Divergence

$$\text{KL}(p||q) = \sum p(x) \cdot \left(\log \left(\frac{p(x)}{q(x)} \right) \right)$$

High KL Divergence



Low KL Divergence

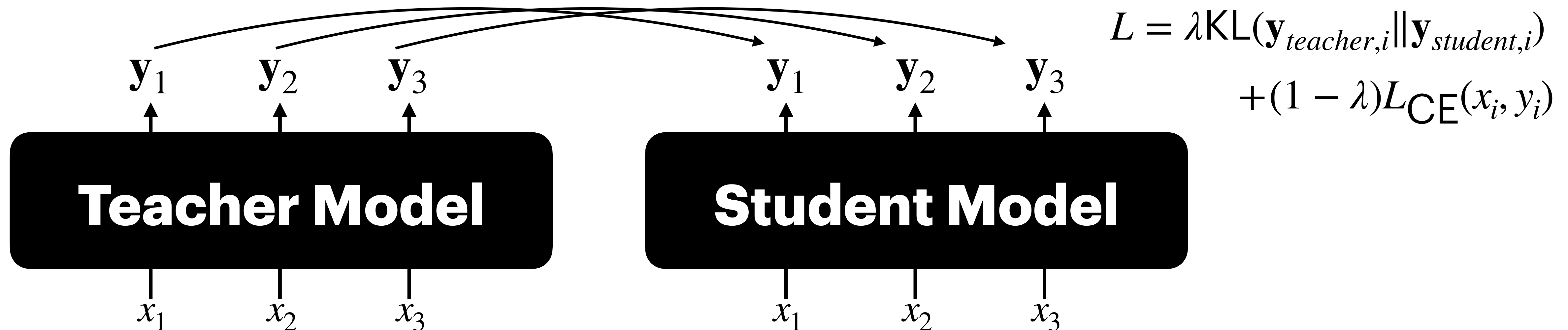


Intuition: how much information is lost when trying to approximate p with q ?

KL Divergence is not symmetric! $\text{KL}(p||q) \neq \text{KL}(q||p)$

Knowledge Distillation

- KL divergence has some cool applications, such as **knowledge distillation**.
- We train a smaller model (the **student model**) to approximate the output distribution of a (frozen) better language model (the **teacher model**) by minimizing the student's KL from the larger model's output distribution on a given token:



- Prior work had simply trained the student on the teacher's output text, but we've since learned that a lot of LM knowledge is not verbalized.

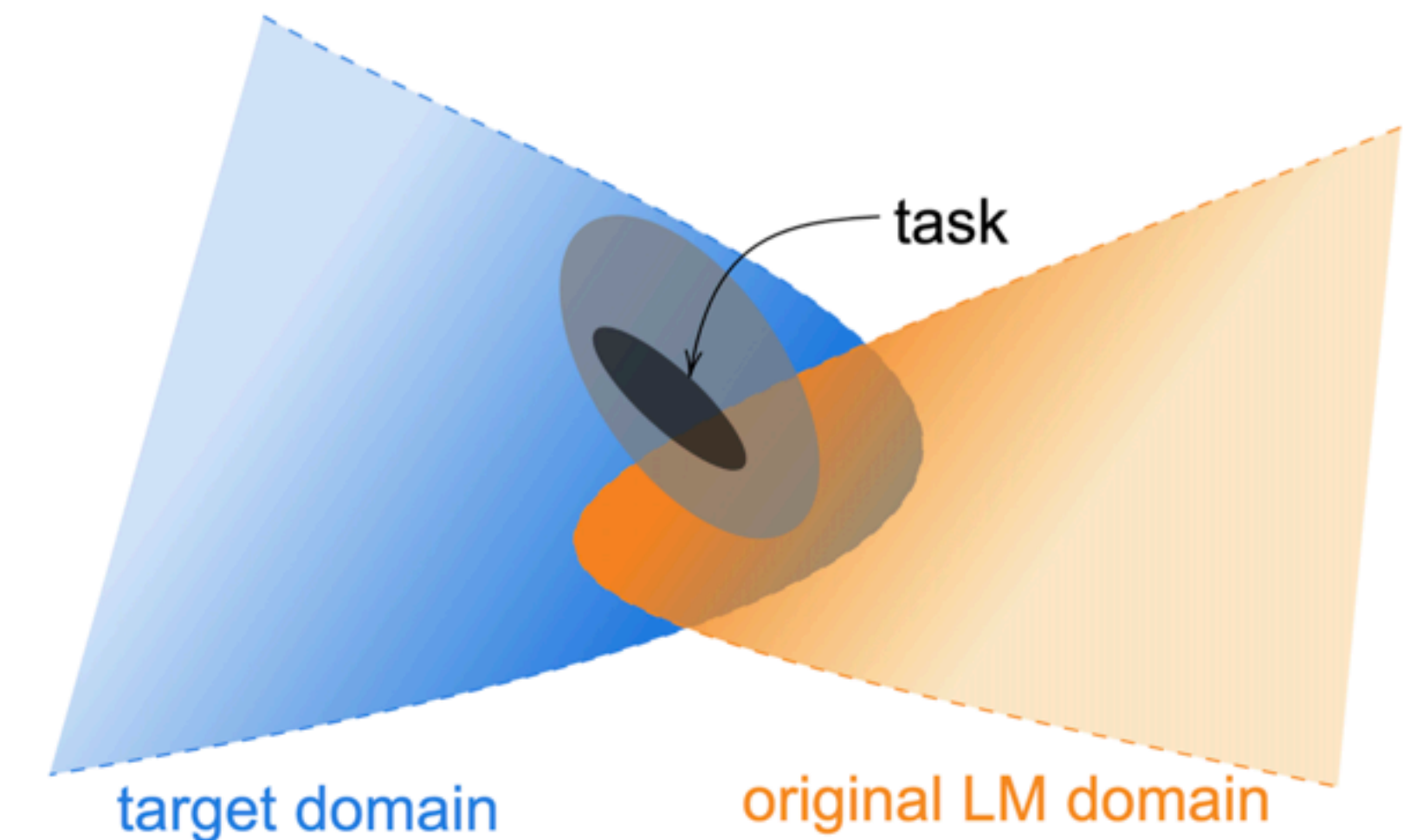
Continued Pre-training

Much like fine-tuning, but for a specific **domain**, not a specific *task*.

Here, the data is usually unlabeled. Given our pretrained model's parameters θ_p , we're just doing language modeling on a specific domain, like medical texts.

$$\theta_0 = \theta_p$$

$$\theta_i = \theta_{i-1} - \alpha \nabla_{\theta} \mathbb{E}_b[-\log p_{\theta}(y_i | \mathbf{x}_i)]$$



Here, y is the correct next token given context $\mathbf{x}$, rather than a class label.

Domain	Task	RoBERTa	Additional Pretraining Phases		
			DAPT	TAPT	DAPT + TAPT
BioMed	CHEMPROT	81.9 _{1.0}	84.2 _{0.2}	82.6 _{0.4}	84.4 _{0.4}
	[†] RCT	87.2 _{0.1}	87.6 _{0.1}	87.7 _{0.1}	87.8 _{0.1}
CS	ACL-ARC	63.0 _{5.8}	75.4 _{2.5}	67.4 _{1.8}	75.6 _{3.8}
	SciERC	77.3 _{1.9}	80.8 _{1.5}	79.3 _{1.5}	81.3 _{1.8}
NEWS	HYPERPARTISAN	86.6 _{0.9}	88.2 _{5.9}	90.4 _{5.2}	90.0 _{6.6}
	[†] AGNEWS	93.9 _{0.2}	93.9 _{0.2}	94.5 _{0.1}	94.6 _{0.1}
REVIEWS	[†] HELPFULNESS	65.1 _{3.4}	66.5 _{1.4}	68.5 _{1.9}	68.7 _{1.8}
	[†] IMDB	95.0 _{0.2}	95.4 _{0.1}	95.5 _{0.1}	95.6 _{0.1}

[Gururangan et al., 2020]

General-purpose Fine-tuning?

- Is there a way to fine-tune such that the model gets better at pretty much everything?
- How can we teach language models to follow instructions more closely?

At what U.S. university was a woman first awarded a Ph.D.?

A. Vassar College B. Harvard University C. University of Pennsylvania D. Cornell University

//

Where can I find information on how to build a house from scratch?

Where can I find information on how to build a house from scratch?
Where can I find information on

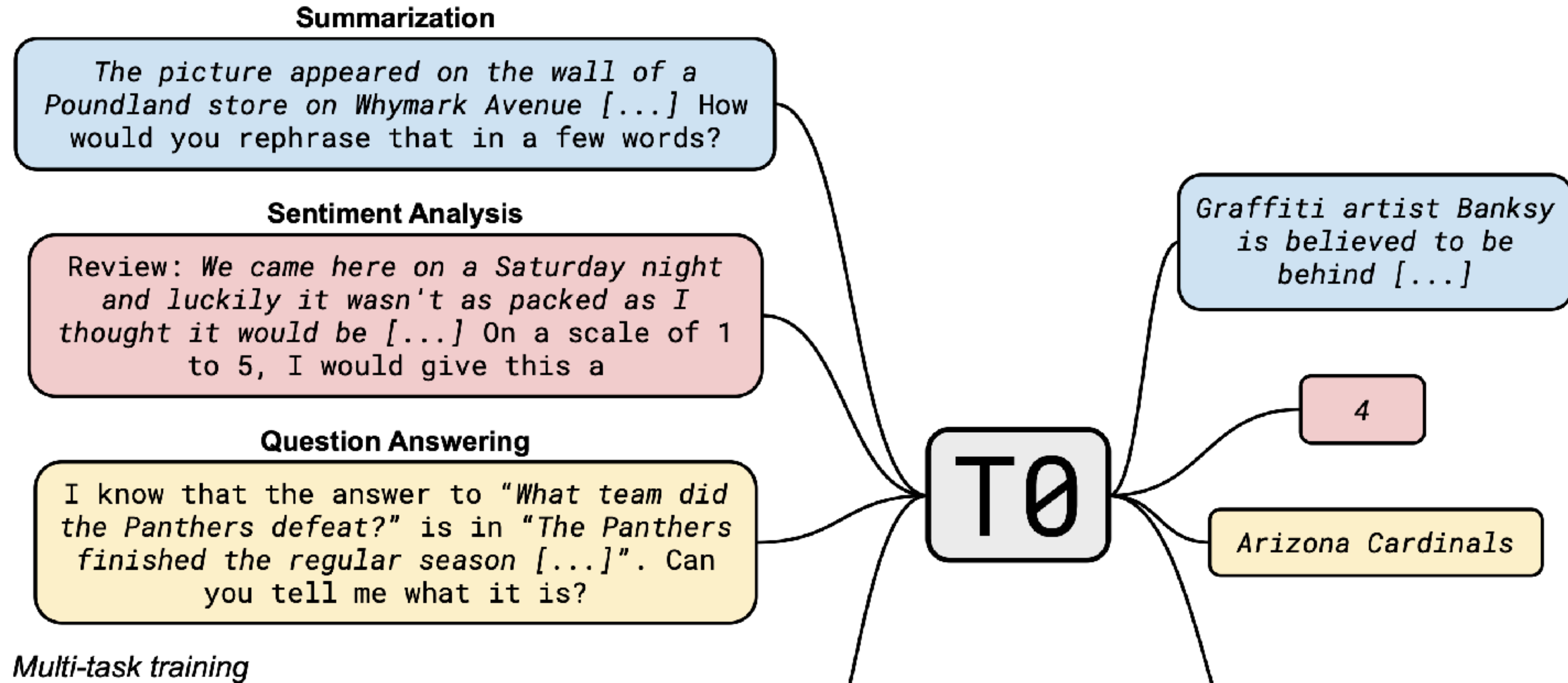
//

Please rewrite the following passage in German: "What is the law of large numbers?" (you can assume that the person you are writing to is familiar with the concept of probability and statistics)

//

Instruction Tuning

[Sanh et al., 2021]



Flan

Finetuning tasks

TO-SF

Commonsense reasoning
Question generation
Closed-book QA
Adversarial QA
Extractive QA
Title/context generation
Topic classification
Struct-to-text
...

*55 Datasets, 14 Categories,
193 Tasks*

Muffin

Natural language inference	Closed-book QA
Code instruction gen.	Conversational QA
Program synthesis	Code repair
Dialog context generation	...

69 Datasets, 27 Categories, 80 Tasks

CoT (Reasoning)

Arithmetic reasoning	Explanation generation
Commonsense Reasoning	Sentence composition
Implicit reasoning	...

9 Datasets, 1 Category, 9 Tasks

Natural Instructions v2

Cause effect classification
Commonsense reasoning
Named entity recognition
Toxic language detection
Question answering
Question generation
Program execution
Text categorization
...

*372 Datasets, 108 Categories,
1554 Tasks*

- ❖ A **Dataset** is an original data source (e.g. SQuAD).
- ❖ A **Task Category** is unique task setup (e.g. the SQuAD dataset is configurable for multiple task categories such as extractive question answering, query generation, and context generation).
- ❖ A **Task** is a unique <dataset, task category> pair, with any number of templates which preserve the task category (e.g. query generation on the SQuAD dataset.)

Held-out tasks

MMLU

Abstract algebra	Sociology
College medicine	Philosophy
Professional law	...

57 tasks

BBH

Boolean expressions	Navigate
Tracking shuffled objects	Word sorting
Dyck languages	...

27 tasks

TyDiQA

Information
seeking QA

8 languages

MGSM

Grade school
math problems

10 languages

1800 tasks!

Chung et al. [2022] train
a model called Flan-PaLM:
540B parameters

Flan-PaLM is pre-trained
first, then fine-tuned on
the many tasks in Flan

Flan-PaLM

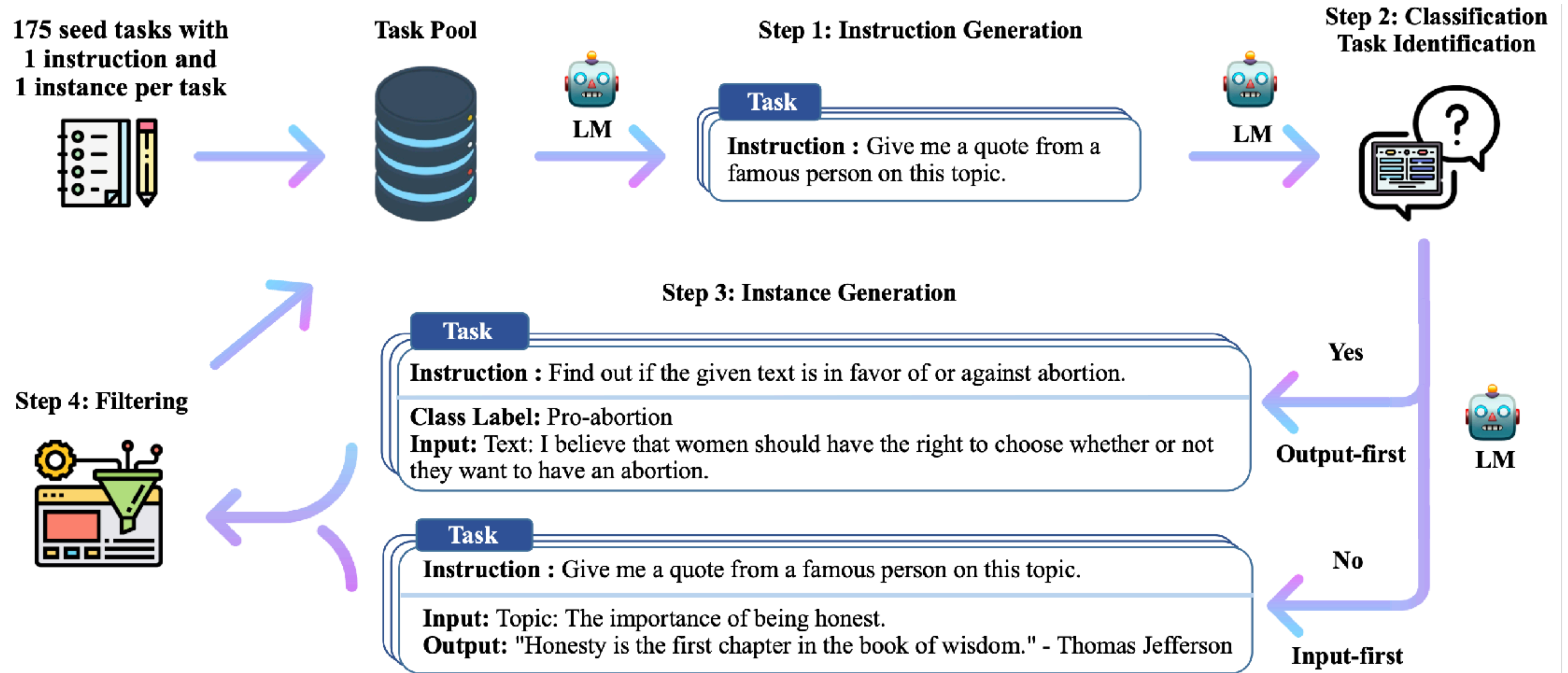
Performance on MMLU [**Hendrycks et al., 2020**]: 57 multiple-choice exam subjects

-	Random	25.0
-	Average human rater	34.5
May 2020	GPT-3 5-shot	43.9
Mar. 2022	Chinchilla 5-shot	67.6
Apr. 2022	PaLM 5-shot	69.3
Oct. 2022	Flan-PaLM 5-shot	72.2
	Flan-PaLM 5-shot: CoT + SC	75.2
-	Average human expert	89.8

Flan

Stephanie - Can you finalize the attached and have it signed. I need to initial it, but it needs to be signed by Brad Richter. Thanks. Write a subject line for this email.	Ronald Chisholm LOI
Ahold to Sell Spain Operations to Permira (AP) AP - The Dutch supermarket retailer Ahold, seeking to streamline global operations and reduce debt, said Sunday it will sell its holdings in Spain to Permira Funds for about \$36.849 billion. What is this text about? OPTIONS: - World - Sports - Business - Science/Tech	Business
Write highlights for this article: Sauntering down leafy avenues past typical Dutch step-gabled buildings, carpets of crocuses, the occasional cyclist whooshing quietly by and the sun bouncing off the canal, I can think of few more pleasant places to spend a spring weekend than The Hague. [...]The most prized Dutch colony was apparently run from The Hague (aka 'the Widow of Indonesia'). For summer: the excellent Grapes & Olives and the Café de Bieb on Veenkade will have boats on a newly renovated bit of canal, with music and boat trips planned, including a tour at the Kompaan brewery (I can vouch for the product). My art gorging finishes at the Panorama, an extraordinary, 14 ft-high circular panorama of The Hague's seaside resort, Scheveningen, painted by Hendrik Mesdag. I take a 15-minute tram to the place itself and gorge on scallops and sole at Catch by Simonis on the harbour front. The Hague has netted a new fan. British Airways (0844 493 0787, www.ba.com) flies to Rotterdam Den Haag (half-hour taxi) from £100 return. For tours with Remco Dörr, email remco.dorr@hotmail.com. Double rooms at the Carlton Ambassador (00 31 70 363 0363, www.carlton.nl/ambassador) cost from £84, room only. More information on travel in The Netherlands at www.holland.com.	The least known of the Dutch cities, The Hague was a village until 1806. It owes its growth to Louis Bonaparte, Napoleon's brother, who ruled here. The city has a wealth of art, including Vermeer's 'Girl With A Pearl Earring'
Here is some data about a restaurant: name = Aromi, eatType = coffee shop, food = English, customer rating = 5 out of 5, area = city centre. Write a sentence that includes the following data about a restaurant	In the city centre there is a coffee shop with a customer rating of 5 out of 5 called Aromi which serves English food.

Self-Instruct/Alpaca



We can have a bigger LM generate instruction-following data for instruction-tuning smaller LMs (Alpaca) or itself (Self-Instruct).

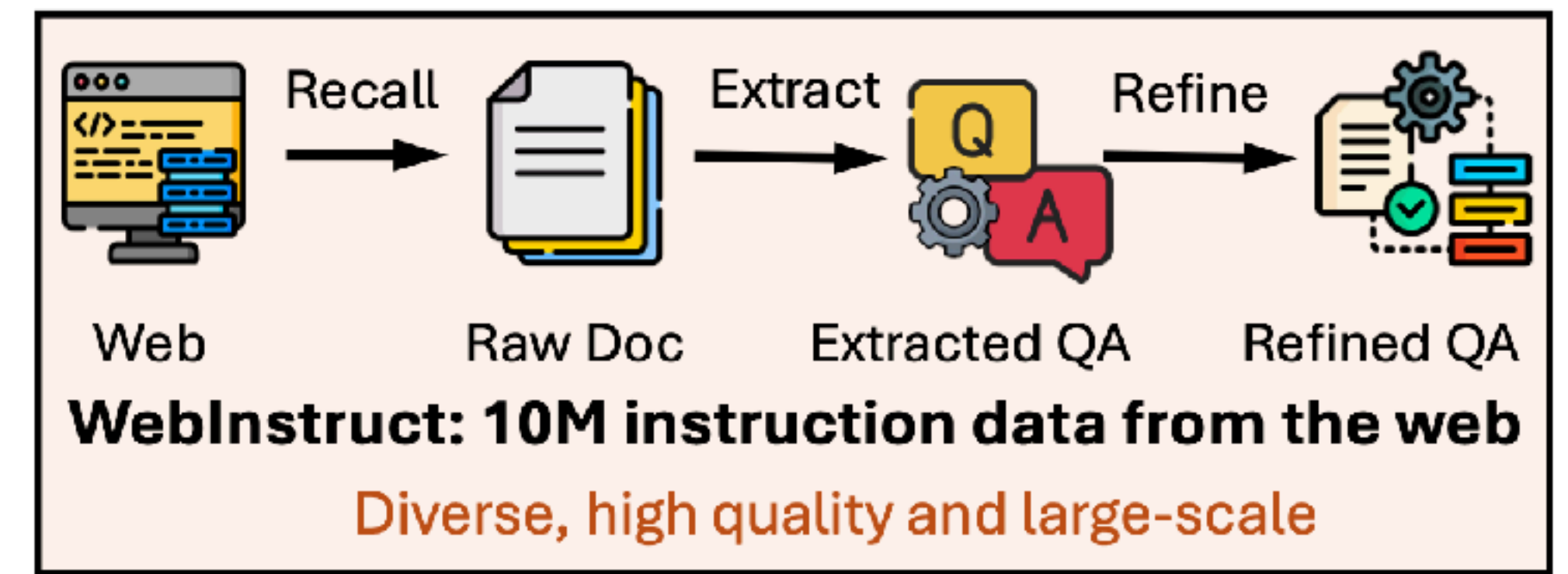
Self-Instruct/Alpaca

Give three tips for staying healthy.	1. Eat a balanced diet and make sure to include plenty of fruits and vegetables. 2. Exercise regularly to keep your body active and strong. 3. Get enough sleep and maintain a consistent sleep schedule.
What does “algorithm” mean?	An algorithm is a step-by-step set of instructions used to solve a problem or achieve a goal. It is a process of breaking down complex tasks into smaller, more manageable steps. Algorithms can be used to find the shortest path from one place to another, sort data, or render data from an immense database in a few seconds.

Instruction Tuning Data in the Wild

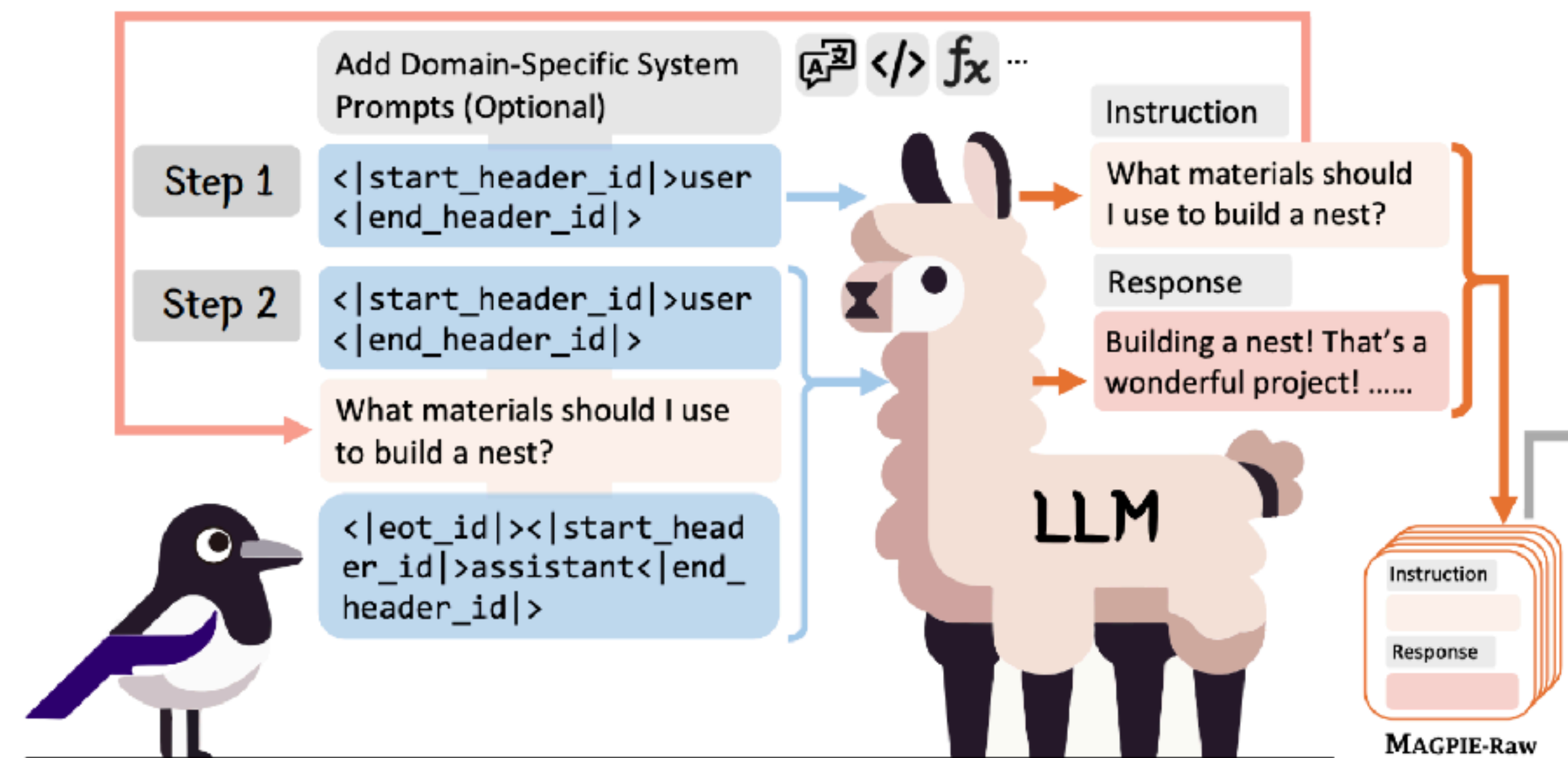
[Yue et al., 2024]

- We can extract instruction-following data from the web
 - (We use LLMs to reformat it.)



- We can generate user prompts using an LLM, have the model respond, filter them and then train on that

[Xu et al., 2024]

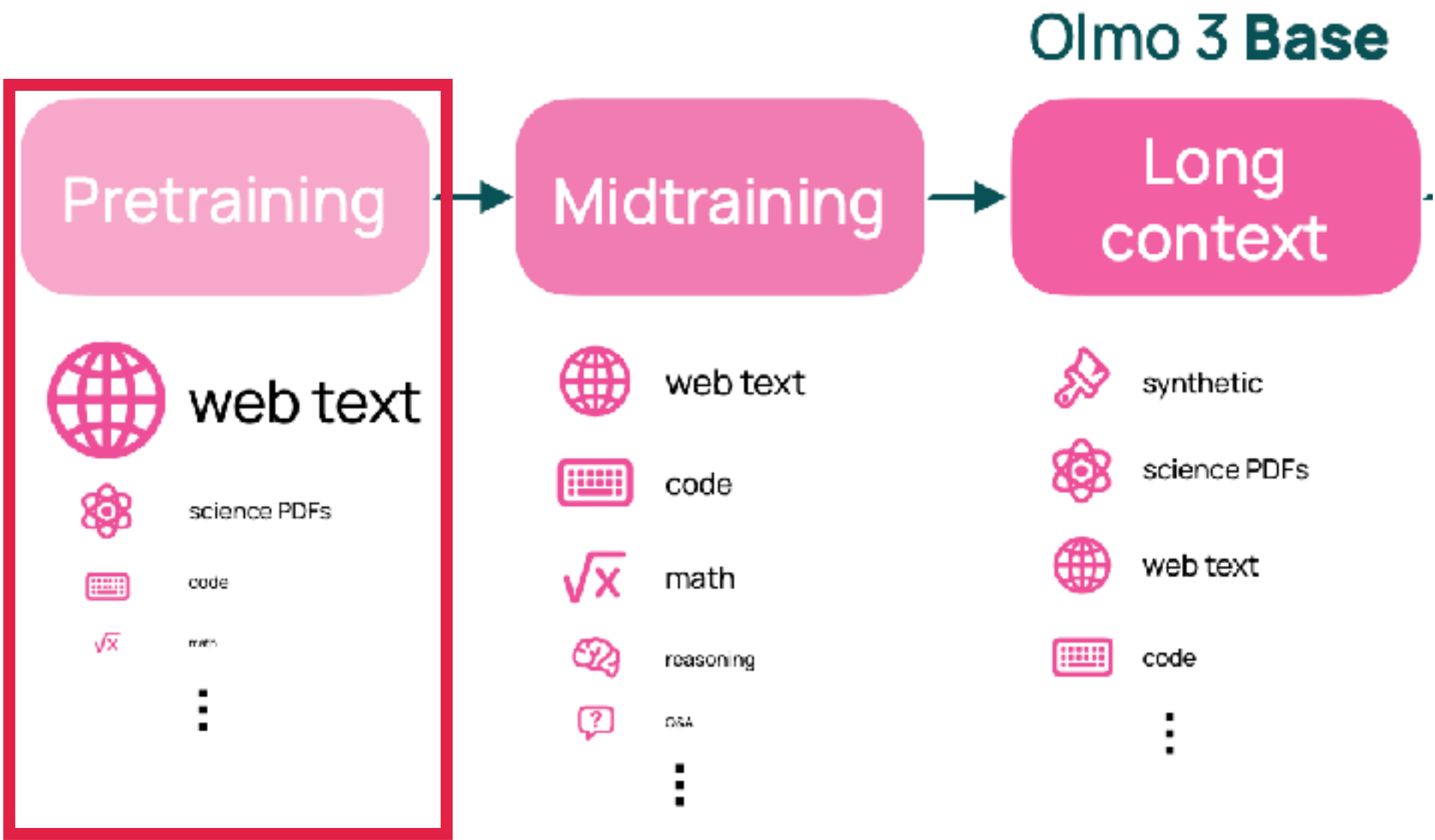


Tulu

	MMLU (factuality)	GSM (reasoning)	BBH (reasoning)	TydiQA (multilinguality)	Codex-Eval (coding)	AlpacaEval (open-ended)	Average
	EM (0-shot)	EM (8-shot, CoT)	EM (3-shot, CoT)	F1 (1-shot, GP)	P@10 (0-shot)	Win % vs Davinci-003	
Vanilla LLaMa 13B	42.3	14.5	39.3	43.2	28.6	-	-
+SuperNI	49.7	4.0	4.5	50.2	12.9	4.2	20.9
+CoT	44.2	40.0	41.9	47.8	23.7	6.0	33.9
+Flan V2	50.6	20.0	40.8	47.2	16.8	3.2	29.8
+Dolly	45.6	18.0	28.4	46.5	31.0	13.7	30.5
+Open Assistant 1	43.3	15.0	39.6	33.4	31.9	58.1	36.9
+Self-instruct	30.4	11.0	30.7	41.3	12.5	5.0	21.8
+Unnatural Instructions	46.4	8.0	33.7	40.9	23.9	8.4	26.9
+Alpaca	45.0	9.5	36.6	31.1	29.9	21.9	29.0
+Code-Alpaca	42.5	13.5	35.6	38.9	34.2	15.8	30.1
+GPT4-Alpaca	46.9	16.5	38.8	23.5	36.6	63.1	37.6
+Baize	43.7	10.0	38.7	33.6	28.7	21.9	29.4
+ShareGPT	49.3	27.0	40.4	30.5	34.1	70.5	42.0
+Human data mix.	50.2	38.5	39.6	47.0	25.0	35.0	39.2
+Human+GPT data mix.	49.3	40.5	43.3	45.6	35.9	56.5	45.2

Case Study: OLMo 3

Pretraining



Source	Type	6T Mix	
		Tokens	Docs
Common Crawl	Web pages	4.51T (76.1%)	3.15B
OLMOCR science PDFs	Academic documents	805B (13.6%)	83.8M
Stack-Edu (Rebalanced)	GitHub code	409B (6.89%)	526M
arXiv	Papers with LaTeX	50.8B (0.86%)	9.10M
FineMath 3+	Math web pages	152B (2.56%)	95.5M
Wikipedia & Wikibooks	Encyclopedic	2.51B (0.04%)	4.24M
Total		5.93T (100%)	3.87B

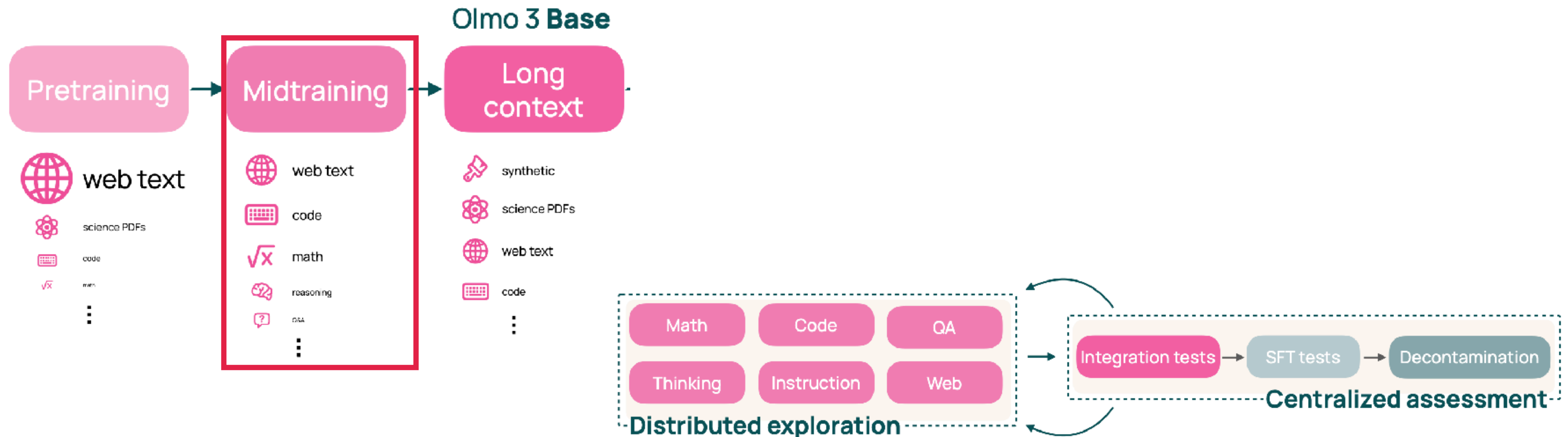
Trained on Dolma 3, a 6T-token dataset.

Mostly web text with some academic papers and math thrown in.

Case Study: OLMo 3

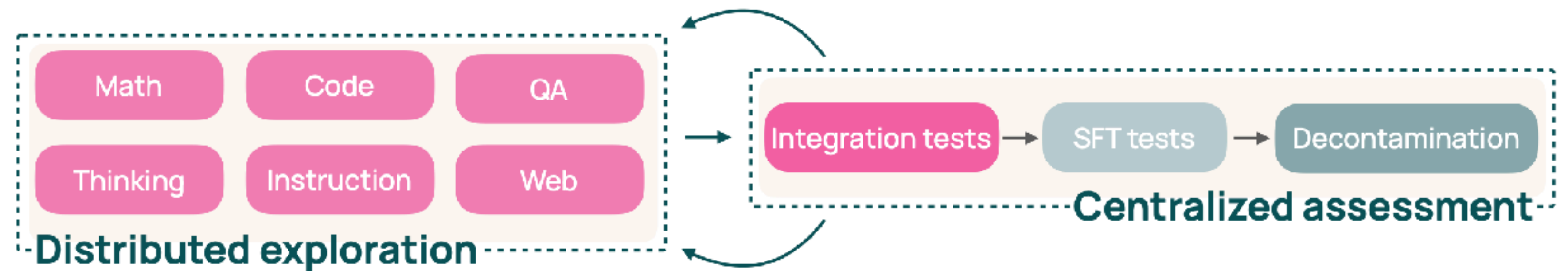
Midtraining

- Models like OLMo 3 do a secondary pre-training step called **midtraining**
- This is like pretraining, but on smaller high-quality curated data (here, 100B tokens)



Midtraining: Data Filtering

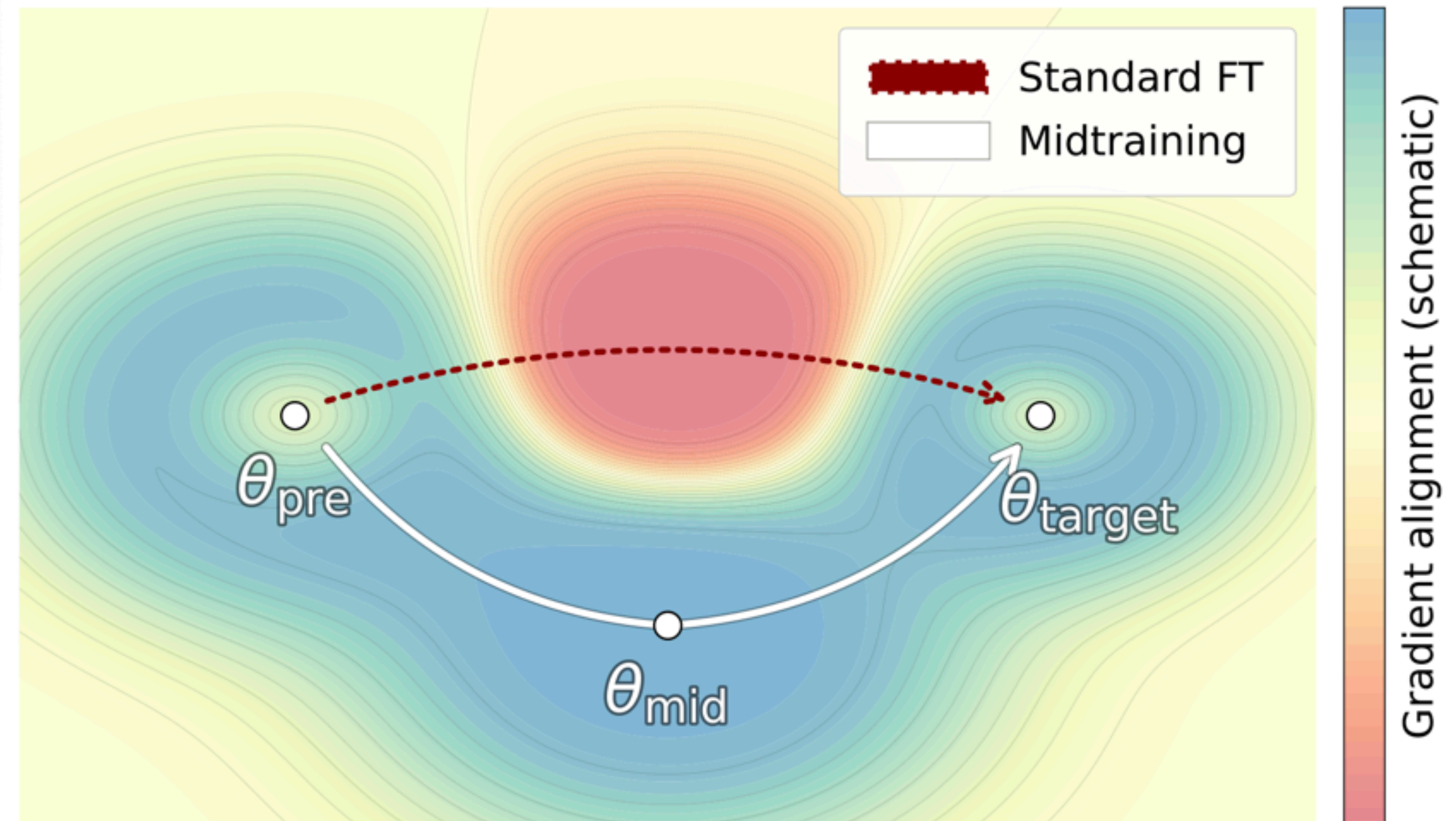
- OLMo 3 uses an iterative data selection and evaluation loop
 - Domain-/task-specific filtering
 - Small-scale continued pre-training/evaluation
 - Adjust filters/data mixture
 - Repeat



What is midtraining actually teaching the model?

[Liu et al., 2026]

- Is midtraining just teaching to the test?
- When is midtraining expected to help in-domain performance, or to make the model forget its pretraining data?
- How does it compare to continued pre-training?

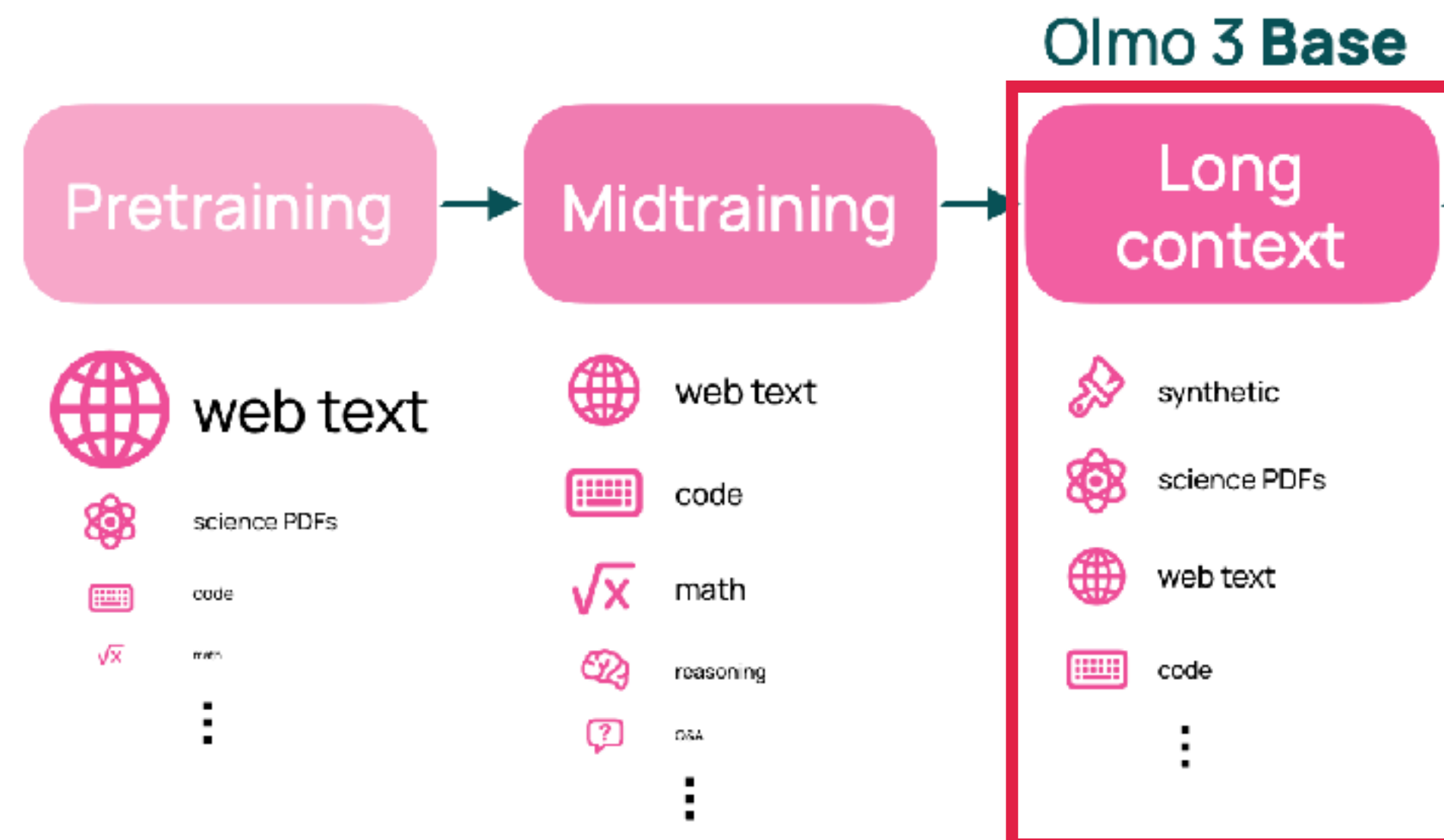


- Midtraining appears to act as a bridge from the pretraining data to specialized domain data
 - Helps reduce catastrophic forgetting
 - Most gains on distant domains from the pretraining corpus

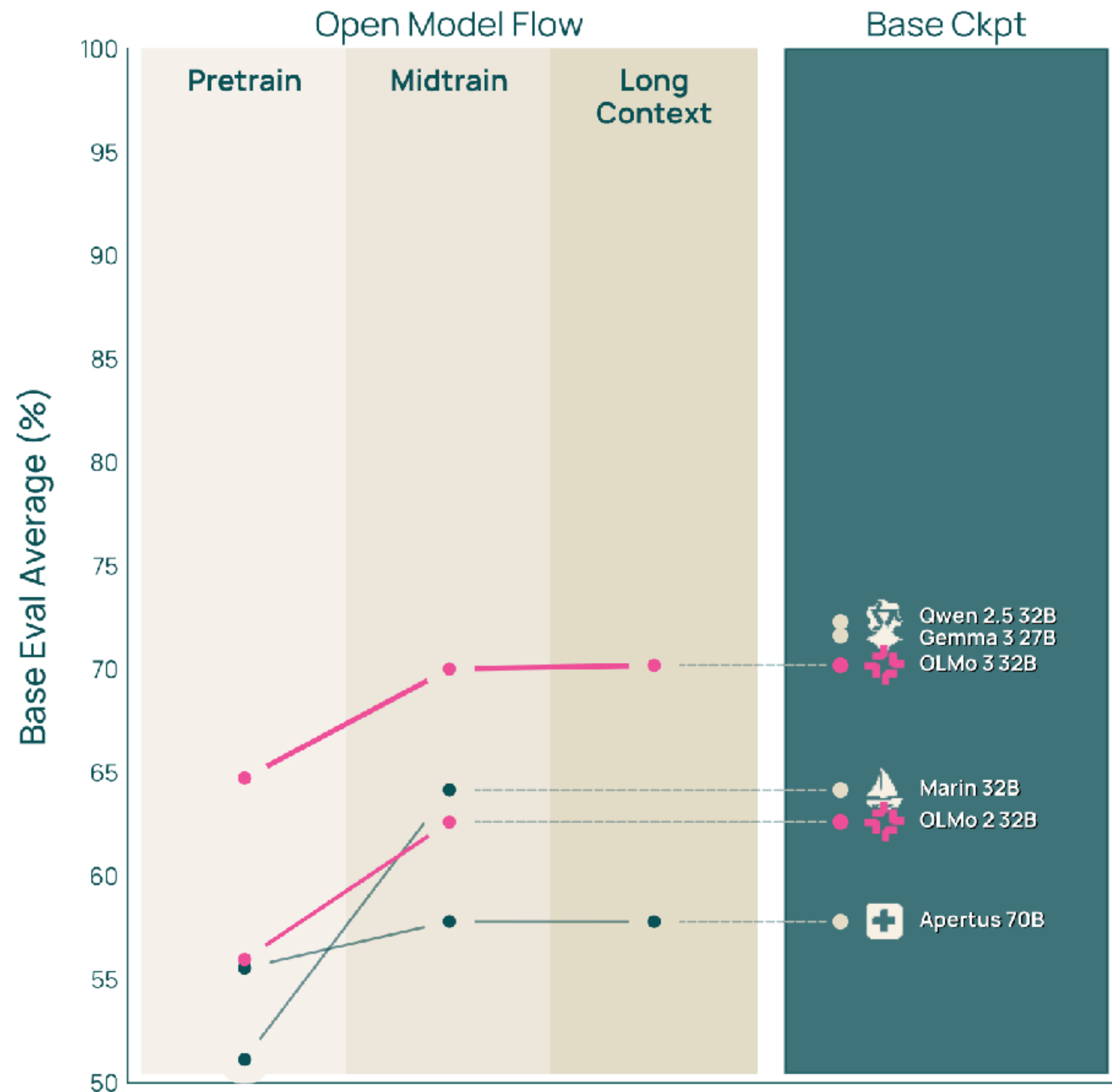
Case Study: OLMo 3

Long Context

- Use data that require following document across many tokens
 - Mostly scientific documents and synthetically generated text
 - Uses RoPE with YaRN to handle longer contexts than were seen during pretraining (see RoPE lecture)



This is our **base model**!



Midtraining helps quite a bit!

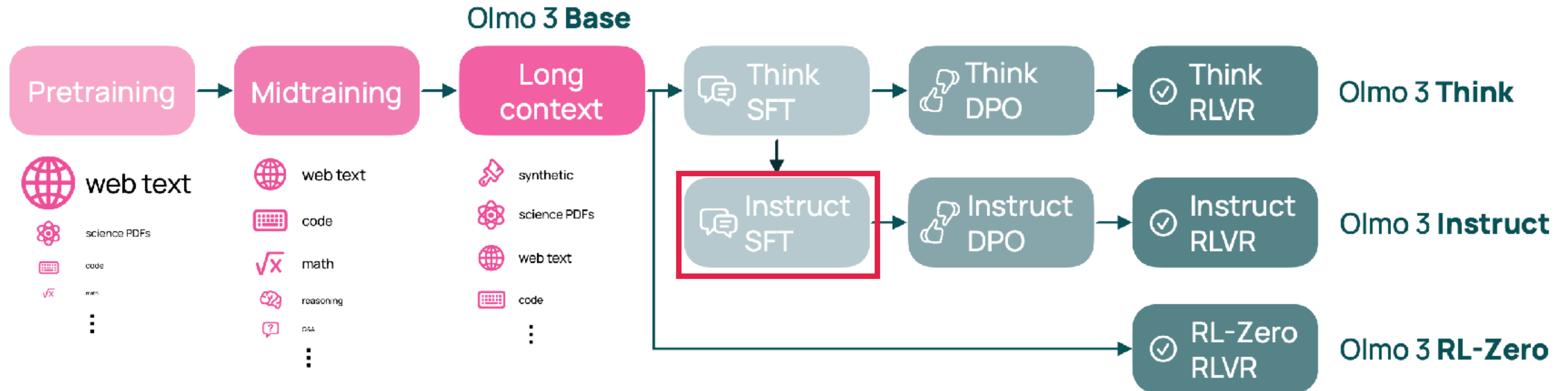
Long-context data does not affect general eval scores much...

Model	RULER (dev suite)				
	4K	8K	16K	32K	65K
7B scale					
Llama 3.1 8B	95.56	92.76	93.13	91.43	86.88
Qwen 2.5 7B	94.63	90.87	88.68	87.26	67.30
IBM Granite 3.3 8B	91.98	85.69	82.70	78.13	67.62
Qwen 3 8B	95.58	94.10	93.78	90.29	-
Xiaomi MiMo 7B	94.33	93.45	92.53	89.28	-
Nemotron Nano 9B	95.31	93.09	91.58	89.01	85.13
Apertus 8B	90.47	82.48	74.43	69.05	59.89
Olmo 3 7B	94.89	91.21	84.14	78.79	67.96

...but it does put OLMo 3 on par with other models w.r.t. long-context reasoning!

From Base to Instruction-following Models

- OLMo 3 comes in many variants! The base model isn't useful for chatting, but there are many variants that are:



- We've seen in this lecture what Instruct SFT looks like. More on the rest of this pipeline in the next few lectures!

Takeaways

- Fine-tuning and midtraining involve applying the same objective as pretraining on a much narrower data distribution.
- We ensure that fine-tuning doesn't stray too far from the pretraining parameters using either regularization or low-rank methods.
- Multi-task fine-tuning and instruction tuning aim to make language models more like general-purpose chat assistants.
- Modern pipelines use midtraining as a bridge between the pretraining and fine-tuning distributions.